

Entangled Photon States, Correlations and Bell Inequality

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This study investigates fundamental quantum phenomena through experiments utilizing entangled photon states generated by Spontaneous Parametric Down-Conversion (SPDC). The research focuses on the characterization of entangled photon pairs and their quantum properties, including fidelity, purity, and non-classical correlations. The fidelity of the reconstructed single-photon quantum state was calculated as $F = 92\%$, with a purity of approximately 89%. For two-photon systems, the reconstructed density matrix exhibited a fidelity of $F = 92\%$ and a purity of 88.98%. Quantum state tomography was employed to confirm high visibility, measured as 97.5% in the horizontal-vertical basis and 92.2% in the diagonal basis. The violation of Bell inequalities, specifically the CHSH inequality, was demonstrated, with $S = 2.634 \pm 0.022$ and strength $n = 28$, for the singlet state and $S = 2.593 \pm 0.021$ and strength $n = 27$ for the triplet state, exceeding the classical bound of $S \leq 2$. Second-order time correlation measurements showed $g^{(2)}(0) = 1.939$ in the non-heralded case and $g^{(2)}(0) = 0.05$ in the heralded case, confirming the particle-like nature of light. These findings have significant implications for advancing quantum communication, information processing, and foundational tests of quantum mechanics, providing a robust framework for optimizing quantum entanglement and correlation techniques.

I. INTRODUCTION

In this series of experiments, we explore fundamental quantum phenomena using entangled photon states generated through Spontaneous Parametric Down-Conversion (SPDC). The laboratory sessions focus on the generation, characterization, and analysis of entangled photon pairs using state-of-the-art quantum optics tools.

Entangled photon states refer to pairs (or groups) of photons whose quantum states are linked in such a way that the state of one photon cannot be described independently of the state of the other(s). This correlation persists even if the photons are separated by large distances, which is a key feature of quantum entanglement.

The first part of the lab investigates the principles and mechanisms underlying SPDC, where a non-linear crystal (BBO) converts a high-energy photon into two lower-energy entangled photons. The resulting quantum states are analyzed through quantum state tomography to reconstruct their density matrices and evaluate key properties, such as purity and fidelity.

Bell inequalities, derived by physicist John Bell in 1964, provide a way to test the validity of local hidden variable theories against the predictions of quantum mechanics. Bell's theorem shows that if nature follows local hidden variables, certain statistical correlations observed in experiments with entangled particles must obey specific inequalities. A prominent example is the CHSH inequality (Clauser, Horne, Shimony, and Holt),

which is a specific form of Bell's inequality used in experiments with two entangled particles. Quantum mechanics predicts that the correlation function S in these experiments can exceed the classical bound of 2, violating the CHSH inequality, which indicates the non-local nature of quantum mechanics. The experimental violation of Bell inequalities has been repeatedly observed, affirming the reality of quantum entanglement and challenging our classical intuitions about space, time, and locality.

The second part of the lab delves into the non-classical correlations and the violation of Bell inequalities, a hallmark of quantum entanglement. By using polarization analysis and examining the CHSH inequality, we assess the degree of entanglement in the prepared Bell states and determine the extent to which our results challenge local hidden variable theories. Additionally, the experiment evaluates photon statistics through measurements of the second-order correlation function, $g^{(2)}(0)$, to confirm the quantum nature of the generated photon streams.

These experiments provide a hands-on understanding of key concepts in quantum optics, such as state preparation, measurement, and statistical analysis of entangled photon pairs, and form a crucial foundation for exploring the applications of quantum mechanics in modern technology and research.

II. ENTANGLED PHOTON STATES

A. Equipment and Theory

The quED system employs *spontaneous parametric down-conversion* (SPDC) to generate polarization-entangled photon pairs. This is a second-order nonlinear

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optical process, where photons from a high-intensity laser, known as the pump beam, spontaneously convert into two lower-energy photons, termed the signal and idler photons.

To produce entangled photon pairs, the quED incorporates two adjacent Type-I nonlinear crystals with optical axes aligned perpendicularly. When the crystals are pumped with light polarized at 45° , the pump photon has an equal probability of converting into either horizontally or vertically polarized photon pairs. Spatial and temporal overlap of the emission from the two crystals ensures the generation of a polarization-entangled state, expressed as:

$$|\Phi\rangle = \frac{1}{\sqrt{2}} (|HH\rangle + e^{i\phi}|VV\rangle),$$

where $|H\rangle$ and $|V\rangle$ represent horizontal and vertical polarizations, respectively, and ϕ is a relative phase determined by the crystal and pump alignment.

Maintaining coherence between emissions is critical for high-quality entanglement. Spatial distinguishability is mitigated by using thin crystals and single-mode fibers, which overlap the emission modes. Temporal coherence is preserved by compensating for group-velocity mismatch and dispersion effects using birefringent crystals. These measures ensure indistinguishability between the two possible emission processes, thus achieving a pure entangled state.

Spontaneous Parametric Down-Conversion (SPDC) is a nonlinear optical process that plays a central role in quantum optics, enabling the generation of entangled photon pairs. It is foundational to numerous quantum experiments, including quantum key distribution, quantum communication, and the violation of Bell inequalities.

SPDC occurs when a high-energy photon from a pump laser interacts with a nonlinear optical medium, such as a birefringent crystal. During this interaction, the pump photon splits into two lower-energy photons, known as the signal and idler photons. The process adheres to the principles of energy and momentum conservation, i.e., the energy of the pump photon equals the sum of the energies of the signal and idler photons. The momentum of the pump photon is distributed between the signal and idler photons, resulting in specific emission angles.

SPDC is classified into two main types, depending on the polarization properties of the emitted photons. In Type I SPDC, both photons have the same polarization, forming concentric cones. In contrast, Type II SPDC generates photons with orthogonal polarizations, producing intersecting emission cones.[1]

This process is fundamental for generating entangled photon pairs, a key resource in quantum technologies. The entangled states produced via SPDC are instrumental for testing the non-classicality of quantum mechanics, implementing quantum key distribution, and exploring advanced quantum protocols.

In the context of the quED system, SPDC is used to produce polarization-entangled photon pairs with high fidelity. The system integrates two nonlinear crystals to create entangled states through coherent spatial overlap of their emissions. Temporal and spatial indistinguishability are ensured through careful alignment, thin crystal design, and compensation techniques using birefringent crystals. Detection and verification of the entangled photons are achieved using polarizers and single-photon detectors.

B. Spontaneous Parametric Down Conversion

1. Photon Count Optimization

Optimizing fiber coupling with a kinematic mount involves fine-tuning the alignment of the fiber coupler to ensure the maximum amount of light is coupled into the fiber. The kinematic mount allows precise angular adjustments in two directions: vertical (up and down) and horizontal (side-to-side). These adjustments are made using two small screws on the mount.

To begin, the fiber coupler is positioned so the incoming light roughly enters the fiber's core. A detector connected to the fiber measures the photon count rate, which acts as feedback for the alignment process. Once the beam is roughly aligned, the screws on the kinematic mount are used to carefully tilt the coupler. Adjusting the vertical screw ensures the light is aligned along the up-down axis, while the horizontal screw ensures the alignment is correct side-to-side. [2]

These adjustments were made very gradually, as even tiny changes can have a big impact on coupling efficiency. As the screws are adjusted, the photon count rate is monitored. If the count increases, the alignment is improving. If it decreases, the adjustment should be reversed or fine-tuned further.

The process is repeated until the photon count rate reaches its highest value, indicating that the light is optimally aligned with the fiber's core and acceptance angle. At this point, the coupling is optimized, and the kinematic mount is secured to prevent any movement that could disrupt the alignment.

2. SPDC Characteristics

After the photon count rate has been optimized, the next step involves systematically recording the coincidence count rate as a function of the laser diode current. This measurement is crucial for understanding the relationship between the pump laser intensity and the generation of photon pairs via spontaneous parametric down-conversion (SPDC). The laser diode current serves as a direct control parameter for the output power of the pump beam, which in turn influences the photon pair production rate.

The measurement begins by setting the laser diode to a low initial current, just above its lasing threshold, in our case is 26mA, as specified in the system's operational guidelines. The current is then gradually increased in small, controlled steps, and at each step, the coincidence count rate is recorded using the photon detection system. Care is taken to ensure that the system remains stable during this process, and the optical alignment, as well as the polarization states, are periodically verified to maintain consistency.

The recorded data allows for the construction of a curve that characterizes the dependence of the coincidence count rate on the laser diode current. This curve typically exhibits a monotonic increase, reflecting the proportionality between the pump power and the photon pair generation rate. However, at higher current levels, saturation effects or thermal instability in the laser diode may become apparent, leading to deviations from linearity, which is not the case in our measurements.

By analyzing this data, the optimal operating current for the laser diode can be identified, balancing high coincidence rates with stable system performance. This characterization is critical for both experimental precision and the reproducibility of results in quantum optics experiments. The findings also provide insights into the efficiency of the nonlinear crystal and the overall performance of the entanglement generation system.

Figure 1 is the plot of Coincidence count to Current. The data collected is displayed in Table II B 2

The electric field associated with the output photons can be described by [3]:

$$E = \sqrt{\frac{n \cdot P_p}{\epsilon_0 \cdot c}}$$

here E is the electric field amplitude, n is the number density of nonlinear optical elements, P_p is the power of the pump beam, ϵ_0 is the permittivity of free space, and c is the speed of light in vacuum. Notice that the power of laser has a quadratic relation with the laser current, meaning the relation between laser current and coincidence count is linear.

Current(mA)	Coincidence Count(pair/s)
0.0	0
26.1	76
27.6	227
29.2	432
30.8	626
32.8	913
34.1	1099
35.2	1211
36.0	1318
37.6	1427
39.2	1604
40.2	1710
41.2	1820

TABLE I. laser current and corresponding coincidence count

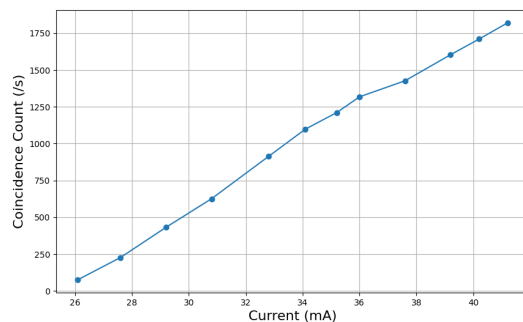


FIG. 1. Coincidence Count to Laser Current

C. Verifying Entanglement

The verification of quantum entanglement in polarization-entangled photon pairs involves a systematic analysis of coincidence rates across different polarization bases. The process begins by inserting polarizing filters into the optical paths of the down-converted photons. Coincidence measurements are initially conducted in the horizontal-vertical polarization basis. In this configuration, the maximum coincidence rates should occur for the combinations of horizontal-horizontal (HH) and vertical-vertical (VV) polarizations, while the minimum coincidence rates should correspond to horizontal-vertical (HV) and vertical-horizontal (VH) polarizations. These results confirm the existence of correlations in the standard polarization basis.

Subsequently, the polarizing filters are rotated to measure the coincidence rates in the diagonal polarization basis. For this basis, the polarization filter positions corresponding to the minimum coincidence rate are set to $+45^\circ/-45^\circ$ (PM) or $-45^\circ/+45^\circ$ (MP), and the minimum coincidence rate, denoted as $N_{\min}$, is recorded. Similarly, the filter positions are adjusted to achieve the maximum coincidence rate in the diagonal basis, and

HH	VV	VH	HV
78	79	2	0
PP	MM	MP	PM
66	82	4	2

TABLE II. Counts after projection to different basis

the corresponding value, $N_{\max}$, is measured.

The quality of the entanglement is then quantified by calculating the visibility of the correlations in the diagonal polarization basis. Visibility is determined using the formula:

$$\text{Visibility} = \frac{N_{\max} - N_{\min}}{N_{\max} + N_{\min}}.$$

A high visibility, typically exceeding 90%, is indicative of strong quantum entanglement. If the visibility is significantly lower than expected, it suggests potential issues such as misalignment in the optical setup or environmental disturbances, which may require troubleshooting and alignment correction.

This procedure provides a rapid and sensitive method for assessing the quality of polarization entanglement and ensures the system is functioning optimally for quantum experiments.

The data collected from experiment is shown in Table II C, also depicted in Figure 2

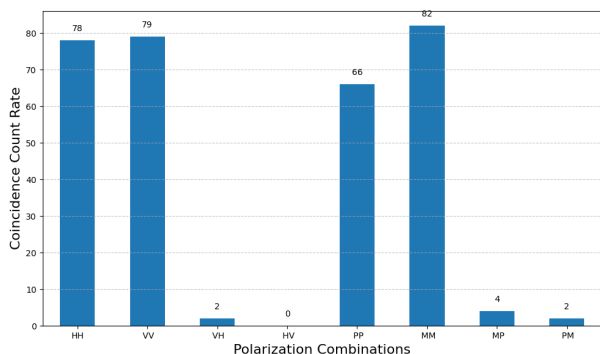


FIG. 2. Coincidence count under different polarization settings

The visibility calculated from this data is 97.5% for H/V basis and 92.2% for P/M basis.

High visibility in the context of polarization correlation measurements is a strong indicator of photon entanglement, but it does not, by itself, provide definitive proof. When visibility approaches or exceeds a certain threshold, typically 70% for two-photon systems,

it suggests that the correlations observed cannot be explained by classical statistical mixtures alone and are consistent with quantum entanglement.

The relationship between high visibility and entanglement is rooted in the behavior of quantum mechanical systems. Entangled photon pairs exhibit correlations in their polarization states that are non-classical in nature. For Entangled state $|\Phi^+\rangle$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle),$$

High visibility in these measurements implies that the observed correlations closely follow the quantum mechanical predictions and exhibit strong coherence. This coherence is incompatible with classical local hidden variable (LHV) models for visibility values exceeding 70%.

To conclusively rule out local hidden variable explanations, it is necessary to perform a test of Bell inequalities, in our case is in the CHSH formalism. Bell inequalities set a strict upper limit on the correlations that can be observed in a system governed by local realism. A violation of these inequalities, often quantified by the parameter $S > 2$ in the CHSH formalism, demonstrates the incompatibility of the observed correlations with any LHV theory. While high visibility is necessary for a significant Bell inequality violation, it is not sufficient alone.

D. Quantum State Tomography

The experimental setup for single photon tomography includes the use of a polarizer and a quarter-wave plate (QWP) placed in one of the beam paths of the quED system.

Once the setup is complete, projection measurements are performed for six fundamental polarization states: horizontal ($|H\rangle$), vertical ($|V\rangle$), diagonal ($|P\rangle$), anti-diagonal ($|M\rangle$), right-circular ($|R\rangle$), and left-circular ($|L\rangle$). The QWP and polarizer are adjusted to specific angles corresponding to each polarization state, and the single photon count rate is measured for each configuration.

The procedure for single qubit quantum state tomography, as described in Section 2.1.2 of the quED-TOM manual, involves setting up the experiment, performing measurements, and reconstructing the quantum state's density matrix. The experimental setup consists of a polarizer and a quarter-wave plate (QWP) placed in one of the beam paths of the quED system. The required components include the quED base unit with quCR and polarizers, along with a single QWP.

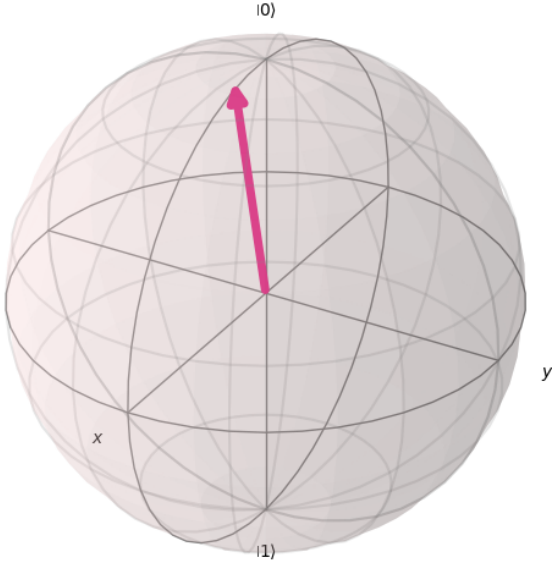


FIG. 3. Bloch vector of the measured state

TABLE IV. Two Qubit Tomography Counts

H	V	P	M	R	L
310	115	188	143	149	185
114	512	162	140	168	117
125	319	228	282	194	168
117	304	214	286	169	160
102	340	190	150	302	293
147	249	171	147	274	168

1. Angle Settings

To perform two-qubit state tomography, one needs to measure each qubit in three mutually unbiased bases: the computational (Z) basis $\{H, V\}$, the diagonal (X) basis $\{P, M\}$, and the circular (Y) basis $\{R, L\}$. Experimentally, these measurements can be implemented using a quarter-wave plate (QWP) and a polarizer. Each of the three bases corresponds to a particular setting of the QWP and HWP before the PBS. The following standard angle settings are commonly employed:

Z-basis (H/V): QWP = 0° , polarizer = $0^\circ/90^\circ$

X-basis (P/M): QWP = 45° , polarizer = $45^\circ/-45^\circ$

Y-basis (R/L): QWP = 45° , polarizer = $90^\circ/0^\circ$

For c_{pr} , the QWP and polarizers should be $45^\circ, 45^\circ$ and $45^\circ, 90^\circ$ for each arm.

2. Measurement Needed to Determine Frequency

To determine the frequency f_{pr} for a particular outcome (p, r) , it is necessary to measure not only the coincidence count N_{pr} corresponding to that outcome, but also to record the total number of detected pairs under the same measurement setting. If the other possible outcomes under the given setting are labeled by (p', r') , then one must measure $N_{p'r'}$ for all such outcomes. With these values in hand, the frequency is given by:

$$f_{pr} = \frac{N_{pr}}{\sum_{p', r'} N_{p'r'}}.$$

Thus, beyond the specific coincidence count N_{pr} , the complete set of coincidence counts for all outcomes, or the total count of detected pairs, is required to accurately determine f_{pr} .

3. The Correlation Tensor Element

The correlation tensor element T_{xy} is defined as the expectation value of the operator $\sigma_x \otimes \sigma_y$, where σ_x and σ_y are the Pauli matrices acting on the first and second qubit respectively. To obtain T_{xy} experimentally, one performs measurements in the $\{P, M\}$ basis (for the x measurement on the first qubit) and in the $\{R, L\}$ basis (for the y measurement on the second qubit).

Assigning eigenvalues $+1$ and -1 to the measurement outcomes corresponding to the eigenstates of σ_x and σ_y , we have:

$$D \leftrightarrow +1, \quad A \leftrightarrow -1, \quad R \leftrightarrow +1, \quad L \leftrightarrow -1.$$

If $f(P, R)$, $f(P, L)$, $f(M, R)$, and $f(M, L)$ denote the joint frequencies of obtaining outcomes (D, R) , (D, L) , (A, R) , and (A, L) respectively, then T_{xy} is given by:

$$T_{xy} = f(P, R) - f(P, L) - f(M, R) + f(M, L).$$

These frequencies are normalized coincidence counts obtained by dividing the raw coincidence counts by the total count of detected pairs. Thus, from the experimentally measured frequencies in the appropriate bases, one can directly compute T_{xy} .

4. Measurement in Tensor Product and Their Influence on Density Matrix

The Pauli matrices are defined as:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

We first compute the tensor product $\sigma_x \otimes \sigma_y$. Using the definition of the Kronecker product:

$$\sigma_x \otimes \sigma_y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Each element of σ_x is multiplied by the entire matrix σ_y :

$$\sigma_x \otimes \sigma_y = \begin{pmatrix} 0 \cdot \sigma_y & 1 \cdot \sigma_y \\ 1 \cdot \sigma_y & 0 \cdot \sigma_y \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}.$$

In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, this can be seen as:

$$|00\rangle \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |01\rangle \equiv \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |10\rangle \equiv \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |11\rangle \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Applying $\sigma_x \otimes \sigma_y$ to these basis states:

$$(\sigma_x \otimes \sigma_y)|00\rangle = i|11\rangle,$$

$$(\sigma_x \otimes \sigma_y)|01\rangle = -i|10\rangle,$$

$$(\sigma_x \otimes \sigma_y)|10\rangle = i|01\rangle,$$

$$(\sigma_x \otimes \sigma_y)|11\rangle = -i|00\rangle.$$

These transformations show how the operator $\sigma_x \otimes \sigma_y$ acts on the individual basis states. Since any two-qubit density matrix ρ can be expressed as:

$$\rho = \sum_{j,k,l,m=0}^1 \rho_{jk,lm} |jk\rangle \langle lm|,$$

linearity allows us to determine how $(\sigma_x \otimes \sigma_y)$ acts on each element $\rho_{jk,lm} |jk\rangle \langle lm|$. Specifically, to find the expectation value:

$$\langle \sigma_x \otimes \sigma_y \rangle = \text{Tr}[(\sigma_x \otimes \sigma_y)\rho],$$

we apply $(\sigma_x \otimes \sigma_y)$ to ρ and then take the trace. The action we computed above on the basis states $|jk\rangle$ determines how each matrix element of ρ is rearranged and multiplied by i or $-i$ factors. In this way, the operator $(\sigma_x \otimes \sigma_y)$ picks out certain off-diagonal coherence elements of ρ and weights them with these complex factors.

5. Density Matrix and Fidelity

Using the data collected for Two Qubit Tomography, which is shown in Appendix IV, we can obtain Density Matrix:

$$\begin{pmatrix} 0.38 & 0.03 & 0.01 & 0.43 \\ 0.03 & 0.09 & -0.01 & 0.01 \\ 0.01 & -0.01 & -0.07 & 0.01 \\ 0.43 & 0.01 & 0.01 & 0.60 \end{pmatrix}$$

The fidelity of the theoretical $|\Phi\rangle$ state is 0.92

The purity is the trace of this reconstructed density matrix is

$$\gamma = \text{Tr}(\rho^2) = 0.3303 + 0.0092 + 0.0052 + 0.5451.$$

Summing these gives:

$$\gamma \approx 0.3303 + 0.0092 + 0.0052 + 0.5451 = 0.8898.$$

Thus, the purity of the given density matrix is approximately 0.89.

III. CORRELATIONS AND BELL INEQUALITY

A. Theory of quED-HBT

The investigation of the particle-like nature of light has been a cornerstone of quantum optics. A key concept in this exploration is photon anti-bunching, which examines whether a beam of light can be characterized as a stream of indivisible particles. This phenomenon is studied using a balanced beam splitter and photon counting detectors, focusing on the anti-coincidence measurement. In such experiments, simultaneous detections by two separate detectors are expected to be highly unlikely if photons truly behave as discrete entities.

Historically, this approach was pioneered by Hanbury-Brown and Twiss, who introduced correlated photon detection as an interferometric technique. Their work led to the development of the second-order time-correlation function, $g^{(2)}(\tau)$, which quantifies the statistical correlation of photon detection events over time. The function is defined as:

$$g^{(2)}(\tau) = \frac{\langle I(t)I(t+\tau) \rangle}{\langle I(t) \rangle \langle I(t+\tau) \rangle},$$

where $I(t)$ is the intensity of the light at time t , τ is the time delay, and $\langle \cdot \rangle$ denotes the time-averaged expectation value. For a truly particle-like source, such as a single-photon emitter, $g^{(2)}(0)$ is expected to approach zero, reflecting the absence of simultaneous detections.

The experimental measurement of $g^{(2)}(0)$ is related to the detection rates of single and coincident events. The probability of detecting two photons simultaneously (a coincidence event) is proportional to the coincidence rate R_c , while the probabilities of detecting individual photons are proportional to the single detection rates R_0 and R_1 . This relationship can be expressed as:

$$g^{(2)}(0) = \frac{R_c}{R_0 \cdot R_1 \cdot T},$$

where R_0 and R_1 are the single count rates of detectors 0 and 1, R_c is the coincidence count rate, and T is the integration time. This equation provides a direct connection between the measured rates and the degree of photon anti-bunching.

In herald case, the relation ship can be expressed as:

$$g^{(2)}(0) = \frac{R_{t12} \cdot R_t}{R_{t1} \cdot R_{t2} \cdot T}$$

Here R_{t12} is the triple coincidence, t here represent the trigger channel.

The experimental setup discussed here involves the use of a beam splitter to analyze a weak source of down-converted photons, allowing for the evaluation of the source's non-classical properties. To complement this experimental work, computational tasks include simulating photon beams using Poisson statistics, estimating photon separations, and calculating distributions of $g^{(2)}(0)$ values. These analyses provide insights into the quantum mechanical properties of light and the methodologies required to probe them effectively.

B. Second-Order Time Correlation

The experimental procedure to measure the second-order correlation function $g^{(2)}(0)$ in a quED-HBT setup, for both non-heralded and heralded cases, involves several key steps:

In the non-heralded setup, the experiment runs continuously as previous one for photons are detected independently by each detector. Coincidences between the detectors are recorded, which are then analyzed to calculate $g^{(2)}(0)$.

For the herald case, connect one single-photon source, i.e. one of the arms, with the input port of a beamsplitter. The beamsplitter directs photons into two separate optical paths, each leading to a single-photon detector. These detectors are capable of recording photon arrival times with high precision.

In the heralded setup, introduce a third detector positioned in a path specifically designed to intercept a fraction of the emission from the photon source. This detector acts as a herald, signaling the presence of a photon in the system. Upon detection of a photon by the heralding detector, the system begins recording coincidences between the two primary detectors. This conditional recording is synchronized with the heralding event, allowing for the calculation of $g^{(2)}(0)$ under the condition that a heralding event has occurred.

TABLE V. Counts to calculate Second-Order Time Correlation for both cases

T	R_0	R_1	R_c
20ns	93791	100625	366
R_{1t}	R_{2t}	R_{12t}	R_t
2875	50	4	104741

For both setups, we ensure that all optical paths are optimized to minimize loss.

The second order time correlation of the not herald case is 1.939. The herald case is 0.05. The experimental data is displayed in Table V

C. Experimental Bell Inequality

In the preparation stage of this portion of the experiment, we once more prepared state $|\phi_s\rangle$ and measured it's visibility. In order to improve experiment efficiency, we utilized the automated system quTools provide. The output of the equipment is displayed in Figure 4

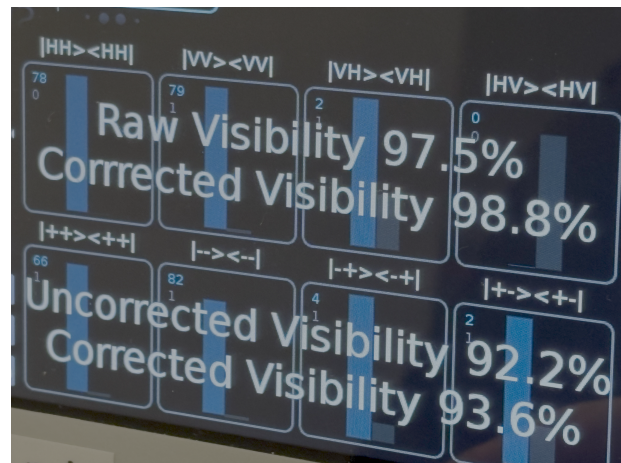


FIG. 4. Output of quED equipment, demonstrating visibility of prepared state

Bell inequality experiment is conducted on this prepared state. The experiment is performed using the quED system to demonstrate the violation of the Clauser-Horne-Shimony-Holt (CHSH) inequality, thereby verifying quantum entanglement [1]. The setup requires a quED base unit equipped with the quCR coincidence counter and two polarizers, which can be either manual or motorized as included in the quED bundle.

The procedure involves measuring coincidence counts for sixteen specific polarizer angle settings, with an integration time of ten seconds for each setting. These measurements are used to calculate the correlation coefficients and the CHSH parameter. To proceed, the polarizers are set to their respective angles, and the coincidence

TABLE VI. Data collected for 16 different polarizer settings for singlet state

Pol 0	Pol 1	Coincidence
H	22.5	470
H	67.5	2104
H	112.5	1873
H	157.5	411
M	22.5	2179
M	67.5	1807
M	112.5	378
M	157.5	477
V	22.5	2074
V	67.5	439
V	112.5	370
V	157.5	1772
P	22.5	435
P	67.5	412
P	112.5	1795
P	157.5	1809

TABLE VII. Data collected for 16 different polarizer settings for triplet state

Pol 0	Pol 1	Coincidence
H	22.5	460
H	67.5	2355
H	112.5	2256
H	157.5	431
M	22.5	563
M	67.5	548
M	112.5	1834
M	157.5	1988
V	22.5	2251
V	67.5	525
V	112.5	370
V	157.5	2132
P	22.5	2251
P	67.5	2264
P	112.5	519
P	157.5	507

counts are recorded for all sixteen configurations. From the recorded counts, the correlation coefficients and the CHSH parameter are computed to evaluate the extent of quantum entanglement.

Data collected for singlet state $|\phi_s\rangle$ are shown in Table VI

The $\langle S \rangle$ is 2.634, dS is 0.022, strength $n_\delta = \frac{S-2}{dS} = 28$

For Triplet state $|\phi_t\rangle$, coincidence count after different projection is shown in Table VII, The $\langle S \rangle$ is 2.593, dS is 0.021, $n = 27$

D. Dependence of the Correlation

We begin by setting one of the polarizers to one of the four canonical polarization configurations: Horizontal (H), +45° diagonal (P), Vertical (V), or -45° diagonal

(M). For each fixed configuration of this first polarizer, we then continuously vary the orientation of the second polarizer across a range of angles, systematically measuring and recording the corresponding coincidence counts between the two detectors. By doing so, we capture a complete angle-dependent coincidence profile for each of the four canonical reference settings. These recorded counts as a function of the second polarizer angle are plotted in Figure 5, where the key CHSH measurement angles, essential for testing violations of local realism, are clearly indicated, H, V, P and M are marked in gray dashed lines while others are marked by gray dashed lines.

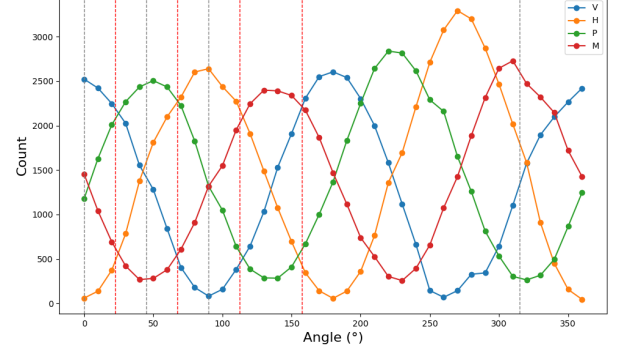


FIG. 5. Output of quED equipment, demonstrating visibility of prepared state

Notice each red dashed line has two intersection point on it while each gray dashed line has one intersection on it. Indicating these are the angles where two sets of projection basis have the same count and one set of projection basis has the same count respectively.

IV. SUMMARY

In this series of experiments, we characterized the properties of a Spontaneous Parametric Down-Conversion (SPDC) source by examining the relationship between pump current and detected count rates. We verified that the generated two-photon state was entangled by measuring the visibility across different polarization bases, thereby confirming the presence of quantum correlations. Both single- and two-photon tomography were performed, allowing us to reconstruct the state vector, determine the density matrix, and evaluate the fidelities of the resulting states.

In the second part of the experiment, we investigated photon correlations and tested the Bell inequalities. The degree of correlation was quantified via second-order time correlation measurements derived from coincidence counts. Contrary to our expectations, the non-heralded time correlation suggested a classical beam. Nonetheless, tests on both singlet and triplet states demonstrated violations of the Bell inequality, reinforcing the nonlocal nature of the generated photon pairs. By examining the

dependence of these correlations on the measurement settings, we gained insights into optimal angles for CHSH measurements.

These findings have significant future implications, as improved control and characterization of entangled photon sources are key to advancing quantum information processing, quantum communication, and fundamental tests of quantum mechanics. The methodologies and results presented in this paper offer a valuable reference point for refining entanglement generation techniques and optimizing Bell tests, ultimately contributing

to more robust and scalable quantum technologies.

Appendix A: Additional Data

The data for Section III.D is too extensive to be displayed on paper, a link to this data is provided: https://github.com/Harrbit/MQST-UCLA/blob/rz_personal_1/qnt_sci_410/labreport3/report/data/cs_data.csv

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